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RESEARCH ARTICLE

DeSFAM: An Adaptive eBPF and AI-Driven Framework for Securing Cloud Containers in Real Time

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ABSTRACT Containerized applications offer lightweight and scalable deployment but remain exposed to security risks due to a shared kernel. We present DeSFAM (Dynamic eBPF-driven Syscall Filtering and Anomaly Mitigation), a real-time security framework that enforces least-privilege syscall usage and detects behavioral anomalies. DeSFAM integrates: 1) hybrid syscall profiling through static analysis and dynamic eBPF tracing; 2) SyscallAD (System call Anomaly Detection), a low-latency anomaly detector combining Variational Autoencoder (VAE) and Isolation Forest (iForest); 3) contextual risk scoring based on MITRE ATT&CK mappings and CVE correlations; and 4) adaptive syscall enforcement using eBPF maps and LSM hooks. Evaluations using the DongTing dataset and real-world CVE attack scenarios show DeSFAM achieves 94% precision, 90% recall, sub-millisecond enforcement latency, and less than 1% performance overhead. DeSFAM effectively blocks privilege escalation, container escape attempts, and syscall injection attacks in modern container environments.

INDEX TERMS Anomaly detection, cloud containers, container security, eBPF monitoring, real-time threat detection, syscall filtering, seccomp hardening.

I. INTRODUCTION

Containerization has become a cornerstone of cloud-native computing due to its efficiency and scalability. Unlike traditional virtual machines, containers share the host OS kernel, leading to reduced overhead but also introducing critical security risks through system calls (syscalls)—the interface between user space and the kernel [1], [2].

Syscalls expose a broad attack surface, making them attractive targets for privilege escalation, container escape, and other exploits [3], [4]. Existing solutions like seccomp profiles offer only limited protection. Static profiles are hard to maintain and may block legitimate operations or fail to detect sophisticated threats [5], [6], [7].

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To overcome these limitations, we propose DeSFAM, Dynamic eBPF-Driven Syscall Filtering and Anomaly Mitigation. DeSFAM provides real-time syscall profiling, deep anomaly detection, and adaptive policy enforcement. It integrates: (i) Hybrid Syscall access listing using static and eBPF-based dynamic analysis; (ii) SyscallAD, a detection engine combining Variational Autoencoder (VAE) and Isolation Forest (iForest) for identifying anomalous syscall sequences; and (iii) Adaptive Policy Management that uses MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) framework mappings, Common Vulnerabilities and Exposures (CVE) correlations, and eBPF LSM hooks for context-aware syscall control.

DeSFAM enforces fine-grained syscall control using eBPF maps and LSM hooks, dynamically blocking high-risk syscalls (e.g., `execve`, `ptrace`) and updating policies at runtime without requiring container restarts. It combines a

hybrid syscall profiling pipeline with SyscallAD, a low-latency anomaly detector based on Variational Autoencoder (VAE) and Isolation Forest. To prioritize threats, DeSFAM integrates contextual risk scoring by correlating syscall behaviors with MITRE ATT&CK tactics and known CVEs. Evaluations using the DongTing dataset and real-world CVE scenarios demonstrate 94% precision, 90% recall, sub-millisecond enforcement latency, and under 1% performance overhead.

Our key contributions are:

- A hybrid profiling pipeline for syscall allowlist generation.
- SyscallAD, a low-latency VAE + iForest anomaly detector.
- Contextual risk scoring using MITRE ATT&CK and CVE correlations.
- Fine-grained syscall enforcement using eBPF maps and LSM hooks.
- Runtime feedback loop for policy auto-hardening based on violations.

The paper is organized as follows: Section II covers syscall filtering and eBPF fundamentals. Section III discusses prior work. Section IV describes DeSFAM's architecture. Section V presents evaluation results, followed by discussion in Section VI and conclusion in Section VII.

II. BACKGROUND

Containerization has fundamentally transformed cloud-native software deployment by offering lightweight, scalable, and portable execution environments. Unlike virtual machines (VMs), which encapsulate a full operating system per instance, containers share the host's OS kernel while providing isolated user spaces via namespaces and control groups (cgroups). This architecture reduces overhead and accelerates deployment, making containers ideal for microservices, CI/CD pipelines, and elastic workloads. However, this shared kernel model introduces unique security risks—particularly at the syscall interface, which governs access to core kernel functionalities.

A. SYSTEM CALL SECURITY IN SHARED-KERNEL ENVIRONMENTS

Syscalls act as the primary interface between user-space applications and kernel resources. In containerized systems, they are critical to process execution, file operations, memory management, and network communication. Unfortunately, their broad capabilities also make them a prime target for attackers. The kernel exposes hundreds of syscalls, many of which are unnecessary for typical containerized workloads, thereby expanding the attack surface [8], [9], [10].

Attackers often exploit misused or vulnerable syscalls to execute privilege escalation, escape containers, or tamper with the host environment. Such threats are exacerbated in multi-tenant cloud deployments, where multiple containers share the same host kernel and thus the same syscall

namespace. Without rigorous syscall control, a compromise in one container can lead to lateral movement or host takeover [11], [12].

B. TYPES OF SYSCALL-BASED THREATS

System calls expose core kernel functions, making them prime targets in containerized environments. Attackers exploit them to escalate privileges, escape isolation, or abuse over-permissive profiles.

Privilege Escalation: Attackers may exploit kernel vulnerabilities, misconfigured Linux capabilities, or incomplete seccomp profiles to gain elevated privileges from within a container [13], [14], [15]. Examples include the use of `setuid()`, `clone()`, or `execve()` in specific sequences to hijack thread execution or bypass namespace restrictions.

Container Escape: Exploits like CVE-2022-0847 (Dirty Pipe) or CVE-2019-5736 (Docker runc) allow malicious processes to bypass container boundaries and interact with the host. These often involve crafting malicious syscall sequences that tamper with read-only memory or escape namespace isolation [7], [16].

Attack Surface Abuse: Many containers have overly broad access to syscalls due to generic or default seccomp profiles. This over-permissiveness can be abused by sophisticated adversaries to chain together syscalls into stealthy, multi-stage attacks.

C. LIMITATIONS OF TRADITIONAL MITIGATION APPROACHES

Security mechanisms such as AppArmor, SELinux, and Seccomp provide syscall filtering and policy enforcement [17], [18]. However, they exhibit several limitations in modern cloud-native environments:

- **Static Policy Inflexibility:** Static profiles do not adapt to new behaviors or threat vectors, often resulting in false positives (over-blocking) or false negatives (missed attacks).
- **Lack of Sequence Awareness:** Rule-based systems operate on individual syscall invocations without considering temporal dependencies, making them blind to multi-step attacks.
- **Scalability Bottlenecks:** As container density increases, static filters become harder to maintain and enforce efficiently at scale.
- **Manual Configuration Overhead:** Policies must be manually written, tested, and maintained for each container workload—an impractical burden in dynamic environments.

These shortcomings highlight the need for dynamic, intelligent, and context-aware syscall security solutions that scale with modern infrastructure [19].

D. THE PROMISE OF EBPF FOR SYSCALL MONITORING

eBPF is a kernel-native technology that allows safe, programmable instrumentation of kernel events, including

syscalls. It enables dynamic attachment of monitoring and filtering logic directly at syscall entry points (e.g., `sys_enter_*`) with minimal overhead [20], [21], [22]. Key benefits of eBPF in the syscall security domain include:

- **Real-Time Enforcement:** Syscalls can be blocked or audited instantly without context switches or restarts.
- **Context-Aware Filtering:** Policies can account for namespaces, cgroup IDs, arguments, or syscall patterns.
- **High Scalability:** Efficient in-kernel maps and trace-points allow enforcement over thousands of containers concurrently.

eBPF's programmable nature makes it an ideal foundation for implementing fine-grained, real-time syscall security tailored to container workloads.

E. SEQUENCE-BASED ANOMALY DETECTION FOR SYSCALLS

Machine learning (ML) models, especially those analyzing syscall sequences, provide a powerful complement to eBPF monitoring. Rather than relying solely on predefined rules, sequence-based models can learn patterns of normal behavior and flag anomalous syscall windows that deviate from expected workflows [23], [24], [25].

Variational Autoencoders (VAE) and Isolation Forests (iForset) are particularly effective for modeling syscall sequences:

- VAEs reconstruct normal behavior from compressed latent space representations. High reconstruction error indicates anomalies.
- Isolation Forests identify statistical outliers based on how easily a sequence can be separated from normal patterns.
- Ensembles of both improve robustness by combining reconstruction-based and tree-based anomaly scoring.

To support training and validation, public datasets such as the DongTing dataset offer syscall sequences labeled as normal or malicious. DeSFAM uses DongTing exclusively, leveraging its 18,966 labeled sequences to train and calibrate its anomaly detection models [26].

F. TOWARD ADAPTIVE, SCALABLE SYSCALL SECURITY

The dynamic nature of container orchestration demands syscall filtering mechanisms that adapt to workload behaviors, incorporate threat intelligence, and scale with minimal human intervention.

DeSFAM addresses these needs by combining:

- Hybrid static-dynamic syscall profiling.
- Ensemble anomaly detection combining deep learning (VAE) and tree-based models (iForest).
- Risk-based policy enforcement using eBPF LSM hooks.
- CVE-matching and MITRE ATT&CK-based risk scoring.
- Auto-hardening mechanisms to refine seccomp profiles at runtime.

This integrated design enables real-time detection, contextual blocking, and long-term syscall surface reduction, setting the stage for resilient and scalable container security in shared-kernel environments.

III. RELATED WORK

This section discusses state-of-the-art approaches categorized by filtering, anomaly detection, provenance tracking, and kernel-hardening, highlighting their strengths and limitations compared to the DeSFAM framework. Table 1 summarizes recent syscall filtering and anomaly detection approaches, highlighting their problem focus, limitations, and core techniques.

A. SYSTEM CALL FILTERING APPROACHES

Confine [31] uses static analysis to construct restrictive seccomp profiles, achieving a 47.6% syscall reduction rate (SRR). However, its static nature prevents adaptation to runtime behaviors, reducing effectiveness against evolving workloads. Similarly, C2C [33] and Prof-gen [36] rely on static techniques that either require manual configuration or omit dynamic tracing, limiting scalability and robustness in real-world deployments.

KubeRosy [29] and Optimus [27] incorporate eBPF for live syscall monitoring and runtime policy updates. KubeRosy leverages Kubernetes CRDs for real-time policy refinement but remains tied to Kubernetes environments. Optimus employs association-based filtering to iteratively adapt syscall profiles, yet its automation is constrained—it requires operator approval for unmatched syscalls and halts profile updates if association analysis fails. Prof-gen, while combining dynamic boot-time tracing with static call graph analysis to generate minimal access lists without source code, lacks support for capturing behavioral shifts during runtime execution.

B. ANOMALY-BASED INTRUSION DETECTION SYSTEMS

Machine learning-based techniques have been widely explored for syscall anomaly detection. The BoSC-based IDS [35] employs Bag-of-System Calls modeling with classifiers like Random Forests, offering reasonable accuracy in small-scale environments. However, frequent retraining requirements hinder scalability and adaptability.

Lightweight models such as the TinyML-based Escape Detector [30] target IoT and edge devices but fall short in handling complex attack vectors in high-performance cloud systems due to limited adaptability and accuracy.

DeSFAM addresses these limitations in Phase 2 via a hybrid anomaly detector combining a Variational Autoencoder (VAE) with an Isolation Forest, trained on the DongTing dataset. This approach enables accurate, low-latency detection of zero-day attacks with no need for retraining, while supporting real-time sliding window inference.

TABLE 1. Comparison of state-of-the-art container security solutions.

Proposed Method	Year	Problem Addressed (Syscall/AD)	Limitations	Tools/Techniques Used
Optimus [27]	2024	Dynamic syscall policy logic	Relies on operator input and halts updates if associations fail	eBPF, Association filtering
Hybrid Syscall Profiling [28]	2024	Static-dynamic syscall profiling gaps	Runtime complexity increases in high-density deployments	Static + dynamic + eBPF
KubeRosy [29]	2024	Lack of real-time syscall enforcement	Restricted to Kubernetes environments	eBPF, LSM, Dynamic updates
TinyML Escape Detector [30]	2024	Container escape in resource-constrained IoT	Model aging and low scalability in cloud	TinyML, Semantic syscall modeling
Confine [31]	2023	Over-permissive syscall policies	False positives, static-only nature	Static analysis, Seccomp
Condo [32]	2023	Kernel permission leakage	Hardware-dependent (TEE required)	Kernel integrity checks, TEE
C2C [33]	2022	Syscall overexposure via configuration files	Manual input needed for precise filtering	Static analysis, YAML filtering
PACED [34]	2022	Cross-container privilege abuse	Overhead from continuous provenance monitoring	eBPF, Provenance tracking
Anomaly-based IDS using BoSC [35]	2021	Anomalous syscall detection	Requires retraining to adapt to new attack patterns	ML, BoSC, Sliding Window
Prof-gen [36]	2021	Static syscall surface minimization	Fails to detect dynamic behavior shifts	Static ELF, Tracing
Sprofiler [37]	2021	Syscall profiling for container isolation	Prone to false positives in noisy environments	Static + dynamic analysis

C. PROVENANCE-BASED DETECTION APPROACHES

Provenance-driven systems like PACED [34] monitor syscall relationships and causality across containers to detect cross-namespace attacks. While effective, they suffer from high overhead due to continuous graph construction and limited scalability.

DeSFAM circumvents this by embedding syscall context (e.g., PID, namespace, timestamps) directly into model features, enabling behavioral awareness without the cost of full provenance tracking.

D. KERNEL-LEVEL SECURITY ENHANCEMENTS

Kernel-layer defenses such as Condo [32] use Trusted Execution Environments (TEEs) to safeguard permission integrity. Though secure, their reliance on specialized hardware and kernel modification makes them unsuitable for large-scale cloud deployments.

In contrast, DeSFAM achieves portability and scalability by utilizing eBPF maps and LSM hooks to enforce syscall policies dynamically, without modifying the kernel or depending on TEEs.

E. COMMON LIMITATIONS OF EXISTING APPROACHES

While there have been meaningful advances in syscall minimization and container security, current solutions face several key limitations:

- Static policies cannot adapt to evolving runtime behaviors.
- Lack of real-time enforcement allows syscall abuse between updates.

- Machine learning approaches require frequent retraining, limiting scalability.
- Hardware dependencies (e.g., TEEs) restrict deployment flexibility.

F. DESFAM: ADVANCING THE STATE OF THE ART

DeSFAM addresses key limitations of prior systems through a unified, multi-phase architecture. It combines hybrid syscall profiling, combining static binary disassembly with dynamic runtime tracing, to generate accurate syscall access lists. Real-time enforcement is achieved via eBPF probes and LSM hooks, enabling low-latency blocking and policy updates without restarting containers. Anomaly detection is handled by SyscallAD, a hybrid model using a Variational Autoencoder and Isolation Forest for precise, retraining-free detection of known and zero-day attacks. DeSFAM also incorporates CVE-aware filtering and MITRE ATT&CK-based risk scoring to restrict vulnerable syscalls dynamically, and applies tiered response logic to enforce fine-grained actions based on threat severity. The framework operates without kernel modification or reliance on orchestration platforms, offering scalable and adaptive syscall security for modern container environments.

IV. DESFAM: DYNAMIC EBPF-DRIVEN SYSCALL FILTERING AND ANOMALY MITIGATION

This section presents the design and implementation of the proposed DeSFAM framework, detailing its architecture, operational phases, and enforcement strategy.

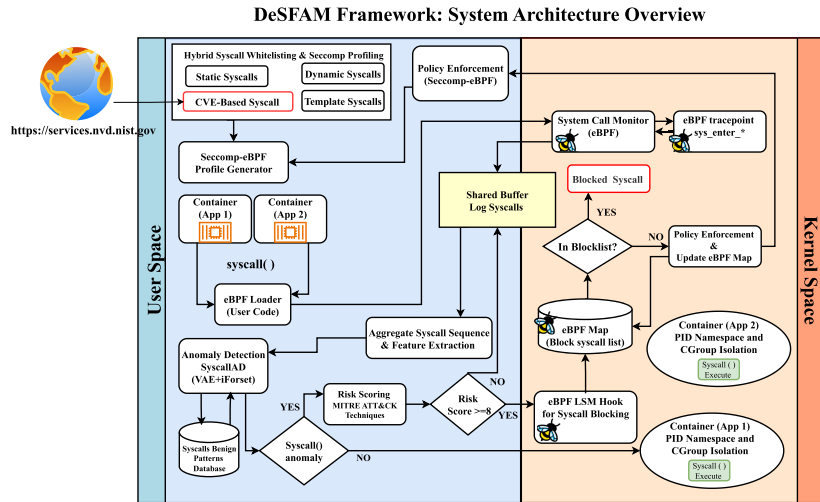


FIGURE 1. DeSFAM architecture showing the interaction between user-space syscall profiling, anomaly detection, and kernel-space enforcement using eBPF and seccomp for adaptive, real-time syscall control.

A. OVERVIEW

DeSFAM (Dynamic eBPF-Driven Syscall Filtering and Anomaly Mitigation) is a lightweight, real-time framework designed to reduce the kernel attack surface and detect anomalous syscall behavior in containerized environments. It enforces least-privilege syscall usage to mitigate threats such as privilege escalation, container escape, and syscall injection. Fig. 1 illustrates the architectural workflow of the framework.

DeSFAM operates in three integrated phases:

- **Hybrid Syscall access listing and Seccomp Profiling:** Combines static call graph extraction and dynamic eBPF tracing to construct a refined syscall access list, pruning unnecessary and CVE-associated syscalls.
- **Real-Time Monitoring and Anomaly Detection (SyscallAD):** Traces runtime syscall sequences and analyzes them using a VAE + Isolation Forest ensemble to detect sequence-level anomalies, including malicious use of permitted syscalls.
- **Adaptive Enforcement and Policy Management:** Enforces blocking, rate-limiting, or logging of high-risk syscalls using eBPF maps and LSM hooks, guided by risk scores from SyscallAD, without disrupting container execution.

By integrating sequence-aware profiling, anomaly detection, and real-time enforcement, DeSFAM offers scalable, adaptive protection suited for modern cloud-native container deployments.

B. DESIGN CONSIDERATIONS

The design of DeSFAM is guided by the need to enforce syscall least privilege while addressing the limitations of static-only filtering, high-overhead provenance tracking, and

retraining-dependent anomaly detectors. Traditional seccomp profiles are either too permissive or brittle, failing to adapt to evolving runtime behavior. DeSFAM overcomes this by combining static and dynamic analysis, enabling accurate and context-aware syscall profiling.

To ensure real-time enforcement without impacting container availability, DeSFAM leverages eBPF probes and LSM hooks, which operate entirely in kernel space and support dynamic policy updates with minimal latency. This design choice avoids the performance cost of continuous provenance graph construction and eliminates the need for kernel modification or container restarts.

For anomaly detection, DeSFAM adopts a hybrid model (VAE + Isolation Forest) that captures behavioral patterns in syscall sequences while avoiding frequent retraining, supporting scalable and robust detection of zero-day attacks. Unlike hardware-bound solutions such as TEEs, DeSFAM remains fully software-based and portable, ensuring compatibility across a wide range of cloud-native deployments.

These considerations low overhead, adaptability, precision, and deployment readiness, form the foundation of the DeSFAM architecture, which makes it suitable for real-world container security at scale.

C. PHASE 1: HYBRID SYSCALL ACCESS LISTING AND SECCOMP PROFILING

Container security hinges on precise syscall control, as containers share the host kernel and pose risks of unauthorized kernel access. Phase 1 of DeSFAM mitigates this by constructing a container-specific syscall access list through a hybrid approach combining static analysis, dynamic profiling, and CVE-based threat intelligence to generate a minimal and accurate seccomp profile.

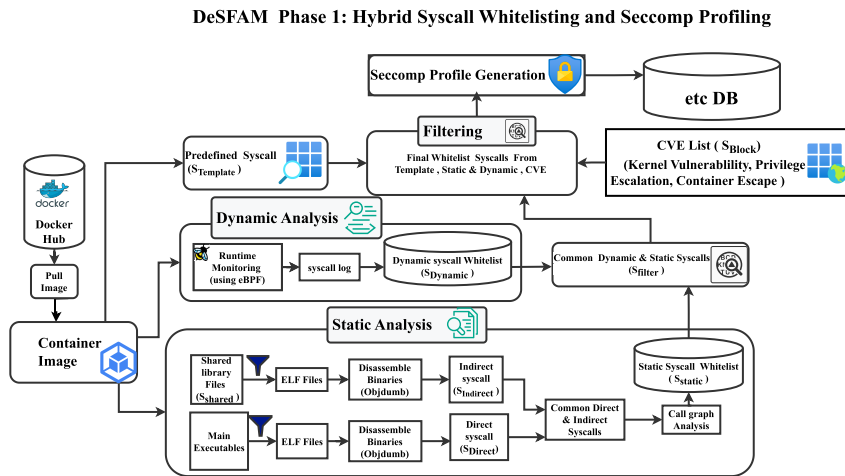


FIGURE 2. DeSFAM Phase 1: Hybrid syscall access listing and seccomp profiling using static analysis, eBPF monitoring, and CVE-informed filtering to generate secure container-specific syscall policies.

1) STATIC ANALYSIS

Static analysis begins by examining container images, including Dockerfiles, binaries, and linked libraries, without executing them. Using tools like `objdump` and `readelf`, ELF binaries are disassembled to extract low-level syscall instructions and libc wrapper functions.

To enhance coverage, a Static Call Graph (SCG) is constructed using NetworkX to trace function-to-syscall mappings across the codebase. The resulting static syscall set, S_{static} , captures all identifiable syscalls reachable from the code's control flow.

- Input: Container image (Dockerfile, ELF binaries)
- Tools: `objdump`, `readelf`, `networkx`
- Process: Disassemble ELF binaries, extract syscall paths via SCG traversal
- Output: Static syscall list (S_{static})

However, static analysis cannot detect syscalls invoked by runtime-resolved libraries or JIT-compiled code. Thus, while S_{static} offers a foundational syscall set, it must be refined through dynamic analysis in subsequent phases.

2) DYNAMIC PROFILING: CAPTURING RUNTIME BEHAVIOR

To address the limitations of static analysis, DeSFAM incorporates dynamic profiling by monitoring live syscall activity during container execution under standardized representative workloads. This phase captures syscalls triggered by runtime conditions including dynamically loaded libraries, environment-specific logic, and dependency interactions, while ensuring comprehensive coverage of typical operational behaviors.

Containers execute under role-specific workloads designed using industry-standard benchmarks that reflect real-world operational profiles:

- Web servers (Nginx):

- wrk with 500-10,000 concurrent connections (static/dynamic content)
- ApacheBench simulating GET/POST requests at 1K-10K req/sec

- Databases (MySQL/Redis/MongoDB):

- SysBench OLTP (Online Transaction Processing) workloads (100-500 threads, 1M+ transactions)
- redis-benchmark at 1M ops/sec across command types
- Custom TPC-C-like transaction scripts with mixed read/write ratios

- Application runtimes (Node.js/Python):

- Locust load tests simulating 50-500 concurrent user flows
- Real-world dataset processing (GB-scale JSON/CSV parsing)

These workloads cover 90th-percentile production traffic patterns, container startup/shutdown sequences, dependency interactions (e.g., web→DB transactions), and failure scenarios (timeouts, connection resets).

During execution, eBPF probes are attached to syscall entry points (`sys_enter_*`) to trace syscall events. The collected metadata includes syscall IDs, process IDs, timestamps, and namespace identifiers, offering detailed insights into operational behavior.

- Input: Container instances under standardized workloads
- Tools: eBPF via `bcc/bpftrace` with kernel-space tracing
- Process:

- 1) Deploy containers with role-specific workload generators
- 2) Attach eBPF probes to syscall entry/exit points
- 3) Log metadata to ring buffers (`perf_event_array`)
- 4) Aggregate syscalls across 3 workload cycles

- Output: Dynamic syscall list ($S_{dynamic}$)

The resulting $S_{dynamic}$ captures runtime-specific syscall patterns, enriching the static access list while maintaining operational realism. This standardized approach ensures policy completeness without overfitting to edge cases.

3) THREAT INTELLIGENCE: CVE-BASED SYSCALL FILTERING

To enhance security awareness, DeSFAM integrates threat intelligence by incorporating CVE data from the National Vulnerability Database (NVD). Using natural language processing (NLP), we extract syscall names (e.g., `ptrace`, `clone`, `execve`) from CVE descriptions to build a list of high-risk syscalls, denoted as $S_{blocked}$.

To construct $S_{blocked}$, we employ an NLP-based parsing engine that analyzes CVE reports retrieved from the NVD via its REST API. A rule-based keyword extraction technique is applied to identify syscall names mentioned in CVE descriptions. We utilize a curated dictionary of Linux syscall names derived from kernel source headers (e.g., `unistd64.h`) to ensure accurate mapping. Furthermore, to avoid overblocking, contextual metadata such as kernel version applicability, affected subsystems, exploit conditions, and argument constraints (e.g., specific device file paths) are extracted from the CVE text using pattern matching and regular expression parsing. CVE entries not applicable to the host kernel (based on `vulnerable_configuration` metadata from the NVD API) are programmatically discarded to ensure threat relevance. This context-aware analysis allows us to differentiate between universally dangerous syscalls and those whose exploitability is limited to specific runtime conditions not present in the target container environment.

- Input: CVE entries with syscall mentions
- Mapping: Rule-based syscall extraction + context filtering
- Output: Blocked syscall list ($S_{blocked}$)

This set is excluded from the union of static and dynamic syscall sets, ensuring vulnerable syscalls are omitted, even if observed during profiling, thus aligning access list generation with real-world exploitability.

4) FINAL ACCESS LIST SYNTHESIS AND POLICY GENERATION

The final access list S_{final} is computed as:

$$S_{final} = ((S_{static} \cup S_{dynamic}) \cap S_{template}) \setminus (S_{blocked} \setminus S_{template}) \quad (1)$$

Here, $S_{template}$ defines role-specific expected syscall behavior for the container (e.g., web server, database, GPU workload), serving as a functional boundary. The template syscalls are derived by combining several sources:

- Official Docker default seccomp profiles;^{1,2}

¹<https://docs.docker.com/engine/security/seccomp/>

²<https://github.com/moby/moby/blob/master/profiles/seccomp/default.json>

- Linux kernel syscall documentation;³
- Community-maintained security rules from the Falco project;⁴
- Empirical syscall traces collected during initial benign container executions using `bpftool` for representative container workloads (e.g., `nginx`, `redis`, `node.js`).⁵

This multi-source approach ensures that rarely invoked but necessary syscalls during container initialization or uncommon execution paths are incorporated into $S_{template}$ while maintaining minimal syscall exposure.

The CVE-based syscall set $S_{blocked}$ is generated by extracting exploit conditions, including kernel versions, device-specific vulnerabilities, or specific syscall argument patterns. This prevents overblocking syscalls that are only vulnerable under specific configurations, not applicable to the container's runtime environment.

To resolve conflicts where CVE-listed syscalls are also deemed essential (present in $S_{template}$), DeSFAM adopts a hybrid security-functional policy, such syscalls are deliberately retained in S_{final} to preserve container functionality but are continuously monitored and tightly controlled at runtime through DeSFAM's adaptive enforcement phases. The resulting access list S_{final} is compiled into a seccomp policy (`seccomp_profile.json`) enforced by the Linux kernel at runtime. Fig 2 and Algorithm 1 summarize this process.

Algorithm 1 Enhanced Hybrid Syscall Profiling

Require: Container image, running instance, CVE database

Ensure: Secure, functional syscall access list

- 1: Extract syscalls via static analysis $\rightarrow S_{static}$
- 2: Profile runtime under workload (dynamic profiling) $\rightarrow S_{dynamic}$
- 3: Parse CVEs with contextual analysis $\rightarrow S_{blocked}$
- 4: Generate $S_{final} = ((S_{static} \cup S_{dynamic}) \cap S_{template}) \setminus (S_{blocked} \setminus S_{template})$
- 5: **return** S_{final}

5) CONTAINER-SPECIFIC POLICY UPLOAD

DeSFAM dynamically loads container-specific syscall policies using `policy_loader`. Each policy reflects the container's behavioral profile from prior analysis and is written into the pinned eBPF map at `/sys/fs/bpf/blocked_syscalls`. Namespace identifiers serve as keys for container isolation.

This design enforces least-privilege syscall access per container at runtime, without restarting instances. Fig 3 shows the upload pipeline and the population of the Syscall map.

³<https://www.kernel.org/doc/html/latest/userspace-api/index.html>

⁴<https://github.com/falcosecurity/rules>

⁵<https://github.com/containerd/common/blob/main/pkg/seccomp/seccomp.json>

```

kubepro@kubepro:~/chapter1$ sudo ./policy_loader container_syscall_policies.json
Policies for container elasticsearch updated successfully.
Policies for container clearlinux updated successfully.
Policies for container intersystems-iris-data-platform updated successfully.
Policies for container orientdb updated successfully.
Policies for container express-gateway updated successfully.
Policies for container redis updated successfully.
Policies for container amazoncorretto updated successfully.
Policies for container heartbeat updated successfully.
Policies for container tomee updated successfully.
Policies for container nats-streaming updated successfully.
Policies for container nuxeo updated successfully.
Policies for container memcached updated successfully.
Policies for container alt updated successfully.

```

(a) Dynamic policy upload using `policy_loader`.

```

kubepro@kubepro:~/bpf-container-policy$ sudo bpftool map dump pinned
/sys/fs/bpf/blocked_syscalls
key: 69 62 6d 2d 64 62 32 2d value: 00 00 00 08 10 00 00 00
key: 61 64 6f 70 74 6f 70 65 value: 00 00 80 00 54 00 00 00
key: 72 65 64 69 73 2d 63 6f value: 00 00 94 ec 15 81 21 00
key: 63 65 6c 65 72 79 2d 63 value: 00 00 00 ec 00 00 00 00
key: 63 6c 6f 6a 75 72 65 2d value: 00 00 00 00 14 00 00 00
key: 69 62 6d 6a 61 76 61 2d value: 00 00 04 ec 14 89 25 00
key: 67 65 6f 6e 65 74 77 6f value: 00 00 00 00 14 00 00 00
key: 73 6f 66 74 77 61 72 65 value: 00 00 00 00 14 00 00 00
key: 63 72 75 78 2d 63 6f 6e value: 00 00 0c ec 14 89 85 00
key: 69 6e 66 6c 75 78 64 62 value: 00 00 14 ec 14 89 25 00
key: 6a 65 74 74 79 2d 63 6f value: 00 00 80 00 54 00 00 00
key: 67 6f 6c 61 6e 67 2d 63 value: 00 00 14 ec 14 89 25 00

```

(b) Container-specific blocked syscall map.

FIGURE 3. DeSFAM Phase 3: (a) Dynamic upload of container-specific syscall policies using `policy_loader`; (b) Verification of blocked syscalls in the pinned eBPF map using `bpftool`.

DeSFAM Phase 2: Real-Time Syscall Monitoring & Anomaly Detection

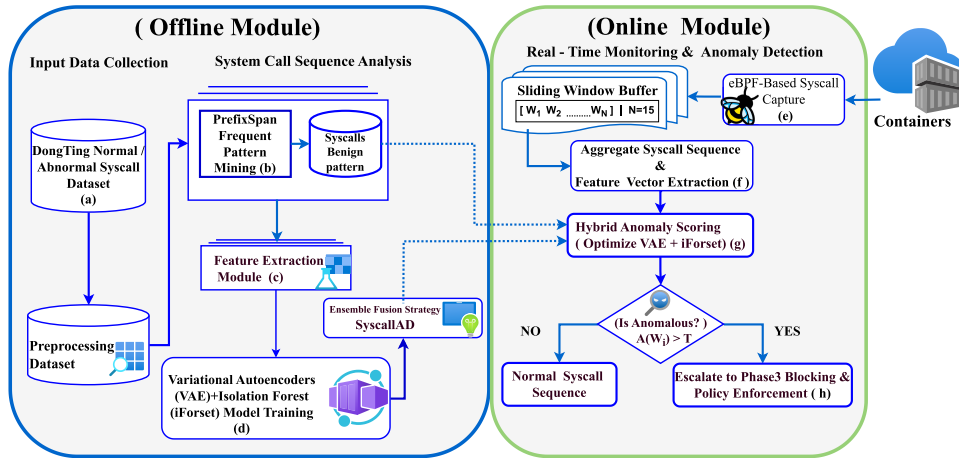


FIGURE 4. DeSFAM Phase 2: Real-time syscall monitoring and anomaly detection using a hybrid VAE + iForest model. The offline module is trained on DongTing sequences, while the online module processes live syscalls via eBPF for anomaly evaluation and policy escalation.

D. PHASE 2: REAL-TIME MONITORING AND SYSCALL ANOMALY DETECTION (SYSCALLAD)

Phase 2 of DeSFAM introduces SyscallAD, a real-time anomaly detection module that combines deep learning and ensemble methods to identify both known and zero-day threats in containerized environments. It monitors syscall sequences using eBPF and detects behavioral deviations linked to privilege escalation, container escape, or syscall injection.

SyscallAD employs a hybrid ensemble comprising a Variational Autoencoder (VAE), which learns latent representations of benign syscall patterns, and an Isolation Forest (iForest), which identifies statistical outliers. This combination enables robust, low-latency detection with minimal false positives and overhead.

1) THE DONGTING DATASET: BEHAVIORAL BASELINES FOR DETECTION

SyscallAD is trained on the DongTing dataset, which comprises 18,966 syscall sequences and over 2.6 million individual syscalls collected from containerized applications. The dataset is split into:

- 6,850 benign sequences: Captured from normal workloads such as web servers and databases executing file I/O (open, read) and memory operations (mmap, futex).
- 12,116 malicious sequences: Generated through simulated attacks, including privilege escalation (setuid), filesystem tampering (unlink), and network intrusion (socket).

Each sequence includes syscall IDs, timestamps, and contextual metadata (e.g., process IDs, namespaces), providing the contextual granularity required for sequence-level modeling. To improve learning reliability, the dataset was cleaned to remove corrupted or incomplete traces and normalized to reduce noise.

Given the inherent class imbalance, more malicious sequences than benign, sampling techniques were applied to prevent model bias. This preprocessing ensures a balanced and high-quality training set, capable of generalizing to real-world scenarios, including zero-day attacks and known exploits such as Dirty COW (CVE-2016-5195).

2) PREFIXSPAN PATTERN MINING: EXTRACTING TRUSTED SYSCALL TEMPLATES

To reduce false positives and improve detection efficiency, DeSFAM integrates PrefixSpan—a sequential pattern mining algorithm—on the benign subset of the DongTing dataset. This identifies frequent syscall subsequences that represent legitimate container behavior.

The resulting Benign Pattern Database is used to:

- Match incoming syscall windows against known-good patterns for early benign classification.
- Bypass deep anomaly scoring for matched sequences, reducing computational overhead.
- Provide semantic traceability for security decisions.

Incorporating PrefixSpan enhances the interpretability and efficiency of SyscallAD by enabling lightweight filtering of known benign activity.

3) FEATURE ENGINEERING: STRUCTURING SYSCALL SEQUENCES FOR DETECTION

SyscallAD applies multi-dimensional feature engineering to convert raw syscall streams into actionable model inputs:

- Sliding Windows: Syscalls are segmented into overlapping windows (length 15, stride 3), capturing both short-term anomalies and extended behavior patterns.
- Categorical Frequencies: Syscalls are mapped to functional categories (e.g., file I/O, memory, network), and their frequency per window forms a behavioral signature.
- Temporal Features: Inter-syscall time intervals (temporal deltas) are summarized (mean, std, max) to detect abnormal timing behaviors.
- Access list Pattern Matching: Each window is cross-checked against benign patterns from PrefixSpan; matches increase confidence in benign classification, reducing false positives.
- Data Augmentation: Malicious sequences are synthetically modified (e.g., insertion or permutation of high-risk syscalls like `clone`, `setuid`) to simulate zero-day scenarios, improving model robustness.

This feature pipeline ensures high-fidelity inputs for the VAE + Isolation Forest ensemble, enabling accurate, scalable, and semantically aware anomaly detection in containerized environments.

4) SYSCALLAD: HYBRID DEEP LEARNING AND ANOMALY DETECTION ENGINE

SyscallAD is a hybrid anomaly detection module designed to identify behavioral deviations in syscall sequences with high precision and adaptability. Unlike traditional point-wise models, SyscallAD combines deep representation learning and statistical outlier detection to capture both complex semantic patterns and rare anomalies.

The system integrates a Variational Autoencoder (VAE) for learning compact latent representations of benign syscall behavior and an Isolation Forest (iForest) for detecting

2025-03-02T13:22:47.799890	Normal behavior detected in container: test-mysql Syscall: futex
2025-03-02T13:22:48.206957	Normal behavior detected in container: test-mysql Syscall: futex
2025-03-02T13:22:48.297612	Normal behavior detected in container: test-redis Syscall: epoll_wait
2025-03-02T13:22:48.350686	Anomaly detected in container: test-redis Syscall: clone
2025-03-02T13:22:49.578179	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:49.586038	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:49.593676	Anomaly detected in container: test-redis Syscall: clone
2025-03-02T13:22:50.844151	Normal behavior detected in container: test-mysql Syscall: futex
2025-03-02T13:22:50.860505	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:50.860505	Normal behavior detected in container: test-redis Syscall: futex
2025-03-02T13:22:51.912785	Normal behavior detected in container: test-mysql Syscall: futex
2025-03-02T13:22:51.970742	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:51.978846	Normal behavior detected in container: test-mysql Syscall: io_getevents
2025-03-02T13:22:52.033732	Normal behavior detected in container: test-mysql Syscall: io_getevents
2025-03-02T13:22:52.233711	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:52.242228	Normal behavior detected in container: test-redis Syscall: wait4
2025-03-02T13:22:53.251668	Anomaly detected in container: test-redis Syscall: ioctl
2025-03-02T13:22:53.324820	Normal behavior detected in container: test-redis Syscall: epoll_wait
2025-03-02T13:22:53.392566	Normal behavior detected in container: test-redis Syscall: wait4

FIGURE 5. SyscallAD output from DeSFAM Phase 2: normal behavior in test-mysql contrasted with anomalous syscall detections in test-redis (e.g., clone, ioctl).

statistical outliers. While the VAE models nonlinear sequence structures and latent dependencies, iForest excels at isolating statistically rare inputs. This hybrid approach enhances generalization to unseen (zero-day) behaviors and reduces false positives.

a: OPTIMIZED VARIATIONAL AUTOENCODER (VAE)

The VAE employs an encoder-decoder architecture (32 $\rightarrow$ 8 $\rightarrow$ 32 neurons) with SELU activations, dropout ($p = 0.2$), and L2 regularization. It learns to reconstruct benign syscall windows by minimizing Mean Squared Reconstruction Error (MSRE). Anomalies are flagged when MSRE exceeds the 99.5th percentile of benign training errors.

b: ISOLATION FOREST (iForest)

iForest isolates anomalies through recursive partitioning. Rare syscall sequences yield shallower isolation trees and higher anomaly scores. A contamination parameter of 2% controls sensitivity, allowing detection of statistically rare behaviors missed by VAE.

Together, the VAE and iForest provide complementary detection capabilities, ensuring robust and efficient identification of both known and unknown syscall threats in real-time.

c: ENSEMBLE FUSION STRATEGY

SyscallAD combines the strengths of VAE and Isolation Forest through a weighted score fusion mechanism to enhance anomaly detection accuracy.

Each syscall window W_i is evaluated by both models:

- $A_{VAE}(W_i)$: Anomaly score based on Mean Squared Reconstruction Error.
- $A_{iForest}(W_i)$: Score derived from isolation depth in the Isolation Forest.

The final score is computed as:

$$A(W_i) = \alpha \cdot A_{VAE}(W_i) + (1 - \alpha) \cdot A_{iForest}(W_i) \quad (2)$$

where $\alpha = 0.7$ prioritizes the VAE's contribution.

The choice of $\alpha = 0.7$ was validated via sensitivity analysis over multiple values ($\alpha \in \{0.0, 0.25, 0.5, 0.7, 0.85, 1.0\}$), using the DongTing dataset. As shown in Fig 6, this setting consistently yielded the best F1-Score and AUC, confirming

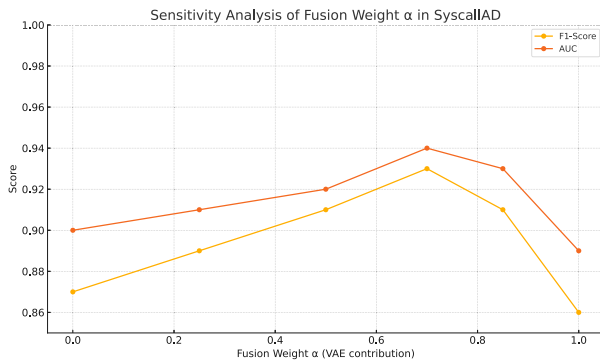


FIGURE 6. Sensitivity analysis of fusion weight α on F1-Score and AUC. Peak performance is observed at $\alpha = 0.7$.

it as the optimal trade-off between structural learning and statistical outlier detection.

Fig 5 presents the anomaly logs from `test-mysql` and `test-redis` containers. Benign syscalls such as `futex`, `epoll_wait`, and `wait4` are correctly classified, while anomalies like `clone` and `ioctl`, indicative of escape or privilege escalation attempts, are effectively flagged. These results demonstrate SyscallAD's ability to distinguish benign from malicious behavior across varied container workloads.

d: DYNAMIC THRESHOLDING: ADAPTIVE ANOMALY BOUNDARY ADJUSTMENT

To accommodate evolving syscall behavior, SyscallAD employs dynamic thresholding based on an Exponential Moving Average (EMA), rather than a fixed cut-off. The initial threshold T_0 is set to the 99.5th percentile of VAE reconstruction errors on benign validation data, allowing tolerance for rare but legitimate variations.

During runtime, the threshold is updated after each syscall window W_i as:

$$T_{t+1} = \max(T_{\min}, \beta T_t + (1 - \beta)A(W_i)) \quad (3)$$

where $\beta = 0.9$ controls update sensitivity, $T_{\min} = 0.5$ enforces a lower bound, and $A(W_i)$ is the ensemble anomaly score.

This adaptive strategy:

- Suppresses false positives from benign behavioral drift.
- Maintains sensitivity to true anomalies and covert attacks.

Final detection is defined as:

$$\text{Flag}(W_i) = \begin{cases} 1, & \text{if } A(W_i) > T \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

By continuously adjusting its decision boundary, SyscallAD improves robustness and precision in dynamic container environments.

```
{
  "timestamp": "2025-01-28T07:34:10.602445Z",
  "syscall_id": 202,
  "pid": 7593,
  "ppid": 8672,
  "container": "kubepro-mysql-1",
  "user": "ib_log_files_g",
  "return_value": 0
}
{
  "timestamp": "2025-01-28T07:34:10.602446Z",
  "syscall_id": 0,
  "pid": 7593,
  "ppid": 0,
  "container": "kubepro-mysql-1",
  "user": "ib_log_files_g",
  "return_value": 0
}
{
  "timestamp": "2025-01-28T07:34:10.602456Z",
  "syscall_id": 202,
  "pid": 7593,
  "ppid": 8672,
  "container": "kubepro-mysql-1",
  "user": "ib_log_files_g",
  "return_value": 0
}
{
  "timestamp": "2025-01-28T07:34:10.612529Z",
  "syscall_id": 0,
  "pid": 7593,
  "ppid": 0,
  "container": "kubepro-mysql-1",
  "user": "ib_log_files_g",
  "return_value": 18446744073709551506
}
{
  "timestamp": "2025-01-28T07:34:10.612544Z",
  "syscall_id": 202,
  "pid": 7593,
  "ppid": 8672,
  "container": "kubepro-mysql-1",
  "user": "ib_log_files_g",
  "return_value": 0
}
```

FIGURE 7. Real-time eBPF syscall log for `kubepro-mysql-1`, showing timestamped syscall events with PID, container name, user, and return value.

5) REAL-TIME MONITORING WITH EBPF: LOW-OVERHEAD CONTAINER-AWARE SYSCALL CAPTURE

DeSFAM utilizes eBPF tracepoints to enable fine-grained, low-overhead syscall surveillance directly at the Linux kernel level. The BPF program hooks into (`raw\_syscalls:sys\_enter`) and (`raw\_syscalls:sys\_exit`), enabling the interception of syscall invocations at their entry and exit points.

To isolate syscall activity from containerized workloads, DeSFAM dynamically maps container PIDs by querying the Docker runtime and injecting them into a `pid_map` BPF hash table. Only syscalls originating from these PIDs are recorded, ensuring container-specific tracing without unnecessary system noise.

Fig 7 captured syscall events are enriched with contextual metadata including:

- syscall ID,
- timestamp (nanosecond resolution),
- process and parent PIDs,
- command name,
- container name (mapped in user space),
- syscall return value.

These are sent from kernel to user space via a `ring_buffer` and logged to per-container files in JSON format. Simultaneously, syscall IDs are appended to a rolling deque (sliding window of size 15) in memory for each container. Once a full window is formed, the syscall sequence is flushed to a structured `.log` file for downstream anomaly detection.

This modular and container-aware monitoring mechanism ensures efficient syscall logging with negligible system overhead, supports live trace-to-sequence conversion, forms the input foundation for SyscallAD's feature extraction and anomaly detection.

6) DevOps INTEGRATION: CONTINUOUS MODEL REFINEMENT

To mitigate false positives and adapt to evolving workloads, DeSFAM integrates a DevOps-driven feedback loop within Phase 2 (SyscallAD). Anomalous syscall sequences, along with metadata (e.g., container name, PID, timestamp), are logged and visualized for analyst review. Security teams can

TABLE 2. Example Syscall-to-MITRE ATT&CK technique mapping.

Syscall	Technique ID	Technique Name	Tactic	Severity (1–10)
ptrace	T1055	Process Injection	Defense Evasion	9
execve	T1059	Command and Scripting Interpreter	Execution	8
clone	T1055.012	Thread Execution Hijacking	Defense Evasion	7
setuid	T1548	Abuse Elevation Control Mechanism	Privilege Escalation	8
mprotect	T1027	Obfuscated Files or Information	Defense Evasion	7
socket	T1071	Application Layer Protocol	Command and Control	6
kill	T1561	Interrupt Inhibitor	Impact	5
unshare	T1611	Escape to Host	Privilege Escalation	9
mount	T1006	Direct Volume Access	Persistence	8
chroot	T1610	Deploy Container	Defense Evasion	8
userns	T1068	Exploitation for Privilege Escalation	Privilege Escalation	9
setns	T1611	Escape to Host	Privilege Escalation	9
bpf	T1547	Boot or Logon Autostart Execution	Persistence	7

Algorithm 2 Real-Time Syscall Anomaly Detection (SyscallAD)

Require: eBPF hook, trained VAE and iForest models, validation reconstruction errors $\mathcal{E}_{val}$

Ensure: Anomaly flag for each syscall window

```

1: Initialize sliding windows  $W_i \leftarrow \emptyset$  for  $i \in \{5, 10, 15\}$ 
2: Load trained VAE and iForest models
3: Compute  $T_0 \leftarrow \text{percentile}(\mathcal{E}_{val}, 99.5)$  {Set initial threshold using 99.5% of benign validation errors}
4: Set threshold  $T \leftarrow T_0$ 
5: while syscall stream is active do
6:   Capture syscall  $s_t$  at time  $t$ 
7:   Append  $s_t$  to each  $W_i$ 
8:   for each full window  $W_i$  do
9:      $f_i \leftarrow \text{extract\_features}(W_i)$ 
10:     $A_{VAE} \leftarrow \text{VAE\_score}(f_i)$ 
11:     $A_{iForest} \leftarrow \text{iForest\_score}(f_i)$ 
12:     $A(W_i) \leftarrow 0.7 \cdot A_{VAE} + 0.3 \cdot A_{iForest}$ 
13:    if  $A(W_i) > T$  then
14:      Flag  $W_i$  as anomalous; escalate to Phase 3
15:    else
16:       $T \leftarrow \max(T_{\min}, \beta \cdot T + (1 - \beta) \cdot A(W_i))$ 
17:    end if
18:    Slide  $W_i$  forward
19:  end for
20: end while

```

annotate detections as true or false positives, with validated false positives appended to the benign training set.

SyscallAD periodically retrains its VAE + iForest ensemble on this augmented dataset, enabling it to learn new benign variations introduced by environment-specific workloads or edge cases. This iterative refinement enhances detection precision and model generalization over time.

The full detection workflow is formalized in Algorithm 2. Syscalls are segmented into sliding windows of multiple lengths (e.g., 5, 10, 15) to balance latency and context depth. Each window is transformed into a feature vector incorporating categorical frequency, timing metrics, and access list matches.

The VAE and iForest models independently score each window; their outputs are fused using a weighted scheme. If the combined score exceeds a dynamic threshold T (initialized at the 99.5th percentile of benign reconstruction errors), the sequence is flagged and escalated to Phase 3. Otherwise, T is updated using an Exponential Moving Average (EMA) to adapt to benign behavioral drift.

By combining real-time detection, adaptive thresholding, and continuous DevOps feedback, SyscallAD maintains high anomaly detection accuracy while minimizing false alarms in dynamic container environments.

E. PHASE 3: ADAPTIVE SYSCALL ENFORCEMENT AND POLICY MANAGEMENT

Adaptive Syscall Enforcement is the third phase of the DeSFAM framework. It focuses on dynamically enforcing syscall policies in containerized environments in real time. This phase integrates risk scoring, contextual analysis, and external threat intelligence to deliver adaptive and precise security responses to minimize syscall abuse in containerized environments with minimal overhead.

1) RISK SCORING MECHANISM

DeSFAM quantifies the threat level of each anomalous syscall sequence W_i using a composite risk score derived from three components:

a: MITRE ATT&CK MAPPING

Each syscall in W_i is mapped to associated adversarial techniques (e.g., T1055 for `ptrace`) based on a predefined lookup (Table 2). This contextual mapping enables reasoning beyond isolated calls to recognize high-level attack patterns.

b: CUMULATIVE SEVERITY

The cumulative severity score aggregates the individual severity scores of syscalls in the sequence:

$$\text{Cumulative Severity} = \sum_{t \in W_i} \text{Severity}(t) \quad (5)$$

TABLE 3. Example risk scores for benign and malicious sequences.

Syscall Sequence (W_i)	Risk Score	Action
open → read → write	4	Log
socket → connect → sendto	5.5	Rate-limit
execve → setuid → mount	8.5	Block
clone → unshare → setns	9	Block
open("/var/log/syslog")	3	Log
ptrace → mprotect → execve (CVE-2021-4034)	13	Block

c: EXPLOIT PATTERN MATCHING

Each sequence W_i is evaluated against known CVE exploit signatures. If a match is found, an additional risk penalty is applied:

$$\mathbb{I}(\text{Exploit Match}) = \begin{cases} 1, & \text{if matched} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

d: FINAL RISK SCORE

The final risk score integrates both severity and exploit intelligence:

$$\text{Risk Score} = \sum_{t \in W_i} \text{Severity}(t) + 5 \cdot \mathbb{I}(\text{Exploit Match}) \quad (7)$$

This multi-dimensional scoring framework enables precise and context-aware enforcement by combining behavioral analysis with real-world threat intelligence.

e: POLICY DECISION LOGIC

Based on the computed risk score, DeSFAM applies a three-tier enforcement strategy:

$$\text{Action} = \begin{cases} \text{Block,} & \text{if Risk Score} \geq 8 \\ \text{Rate-limit,} & \text{if } 5 \leq \text{Risk Score} < 8 \\ \text{Log,} & \text{if Risk Score} < 5 \end{cases} \quad (8)$$

These thresholds are empirically tuned and remain configurable for deployment-specific risk postures. Table 3 presents sample sequences and corresponding actions.

This mechanism enables real-time, context-aware enforcement that adapts to both benign variations and high-risk attack signatures.

2) eBPF-BASED ENFORCEMENT

DeSFAM enforces syscall policies through eBPF programs attached to LSM hooks. Specifically, these programs are loaded and attached using `bpf_program_attach_lsm()`; to intercept system calls at the LSM's `security_syscall` hook. This allows DeSFAM to observe and act on all syscalls issued by containerized processes.

a: LSM HOOK ATTACHMENT

eBPF programs are dynamically attached to the `security_syscall` LSM hook using `bpf_attach_lsm()`:

```
// eBPF Hook Attachment struct bpf_program
*prog = bpf_object__find_program_by_name(obj,
```

```
    ``enforce_policy``);
struct bpf_link *link = bpf_program_
attach_lsm(prog);
```

This enables interception of all syscalls during entry execution. The hook function signature:

```
SEC(``lsm/syscall``)
int BPF_PROG(enforce_policy, int syscall_num,
struct pt_regs *regs) {
    // Enforcement logic here
}
```

Kernel integration ensures all containerized processes are monitored regardless of their namespace context.

b: POLICY MAP STRUCTURE

Syscall actions are stored in a hash-based eBPF map indexed by container identifiers:

```
struct container_key {
    u32 mntns; // Mount namespace ID u32 pidns;
    // PID namespace ID
};

struct bpf_map_def SEC(``maps``) policy_map =
{.type = BPF_MAP_TYPE_HASH, .key_size =
sizeof(struct container_key), .value_size =
sizeof(u64), .max_entries = 144,
};
```

The map is pinned at `/sys/fs/bpf/blocked_syscalls` for runtime updates via `policy_loader`.

c: SYSCALL BLOCKING MECHANISM

The actual blocking of syscalls occurs within the eBPF hook attached to the `security_syscall` LSM. When a monitored syscall is invoked, the `enforce_policy()` function is triggered, and the critical enforcement primitive uses kernel function override: `bpf_override_return(regs, -EPERM)`; // Immediate syscall denial

This workflow implements blocking:

- Retrieve container context:

```
u32 mntns = bpf_get_ns_id(NS_MNT);
u32 pidns = bpf_get_ns_id(NS_PID);
```
 - Query policy map:

```
u64 *action = bpf_map_lookup_elem
(&policy_map, &key);
```
 - Enforce action:
 - BLOCK: Deny via `bpf_override_return(regs, -EPERM)`;
 - RATE_LIMIT: Allow N invocations/minute via atomic counters
 - LOG: Record metadata via ring buffer
- Syscalls not in the map are permitted.

d: SELF-HEALING ACCESS LIST VIA SECCOMP REGENERATION

To ensure long-term policy evolution, DeSFAM incorporates a self-healing mechanism that automatically refines static syscall access lists based on repeated runtime blocks.

```

struct probation_key {
    struct container_key ckey;
    u32 syscall; // Syscall number
};

```

```

SEC("maps") struct bpf_map_def probation_map = {
    .type = BPF_MAP_TYPE_LRU_HASH,
    .key_size = sizeof(struct probation_key),
    .value_size = sizeof(u32), // Block counter.
    .max_entries = 512,
};

```

Blocked syscalls are tracked in-kernel using a Least Recently Used (LRU) cache map called `probation_map`, indexed by the tuple `(mntns, pidns, syscall)`, abstracted in the struct `probation_key`.

If a syscall is blocked three or more times within a configurable window and is not previously annotated as benign, the kernel triggers a user-space daemon via `bpf_send_signal(SIGUSR1)`.

This daemon then:

- Resolves the syscall name using its number (e.g., 105 $\rightarrow$ `setuid`)
- Locates and updates the corresponding container's seccomp profile
- Removes the flagged syscall from the access list
- Reapplies the updated profile to future container instances

This adaptive defense loop ensures that newly deployed containers inherit hardened, exploit-aware syscall policies without requiring restarts or manual review.

e: FORMAL POLICY MODEL

Let S be the set of valid syscalls, and M, N represent mount and PID namespaces. The enforcement policy is defined as:

$$P : S \times M \times N \rightarrow \{0, 1\} \quad (9)$$

$$\text{Decision}(s) = \begin{cases} \text{Allow,} & \text{if } P(s, mntns, pidns) = 1 \\ \text{Block,} & \text{otherwise} \end{cases} \quad (10)$$

This container-aware logic ensures syscall isolation and enforces least-privilege access per namespace. The combined use of eBPF tracepoints and LSM hooks enables fast, layered enforcement with minimal overhead.

f: ALGORITHM: ADAPTIVE SYSCALL ENFORCEMENT

The enforcement procedure in Phase 3 is formalized through Algorithm 3, which provides a structured and replicable workflow for translating risk assessments into real-time syscall policy updates. It encapsulates the integration of risk scoring, exploit pattern matching, and runtime enforcement using eBPF and LSM hooks. This algorithm serves as an implementation blueprint for the DeSFAM framework's adaptive defense layer, supporting both reproducibility and rigorous validation.

DeSFAM Phase 3: Adaptive Syscall Enforcement & Policy Management

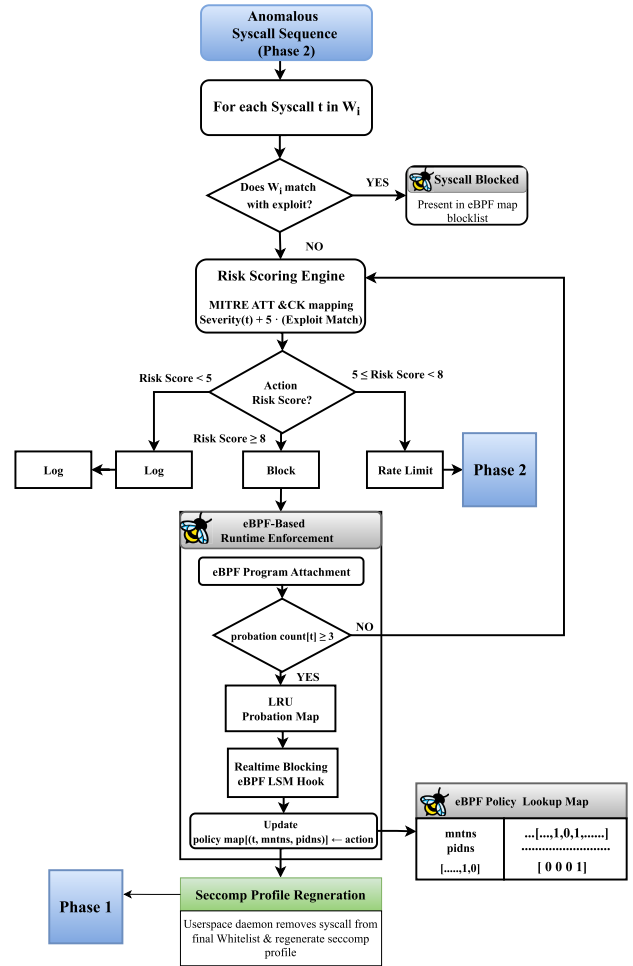


FIGURE 8. DeSFAM Phase 3 – Adaptive enforcement pipeline. Anomalous syscall sequences from Phase 2 are intercepted and acted upon via eBPF and LSM hooks.

3) USE CASE: PRIVILEGE ESCALATION ATTEMPT AND REAL-WORLD CVE DEFENSE

Consider a container running under DeSFAM's restrictive policy designed to block privilege escalation attempts. A typical attack sequence—`clone()`, `setuid()`, `execve()`—is intercepted at the kernel level.

Assume the process operates in mount namespace `mntns = 1001` and PID namespace `pidns = 2001`. Upon invocation of `setuid()`, DeSFAM consults its per-container policy:

```

policy_map[1001, 2001] = [..., 1, 0, 1, ...],

```

where each entry denotes the action for a syscall ID (0 = BLOCK, 1 = ALLOW). Since `setuid()` is mapped to 0, the syscall is denied.

The eBPF LSM hook, `realtime_blocking`, enforces this decision in the kernel, returning `-EPERM` and logging the event for auditing. This ensures immediate, in-kernel

Algorithm 3 Adaptive Syscall Enforcement With Auto-Hardening

Require: Anomalous sequence W_i , MITRE mapping, exploit database, container context ($mntns$, $pidns$), eBPF map $policy_map$

Ensure: Enforcement action: “Block”, “Rate-limit”, or “Log”

```

1: Initialize  $Risk\_Score \leftarrow 0$ 
2: for each syscall  $t$  in  $W_i$  do
3:    $Risk\_Score \leftarrow Risk\_Score + Severity(t)$ 
4: end for
5: if  $W_i$  matches an exploit in the database then
6:    $Risk\_Score \leftarrow Risk\_Score + 5$ 
7: end if
8: if  $Risk\_Score \geq 8$  then
9:    $action \leftarrow$  “Block”
10: else if  $5 \leq Risk\_Score < 8$  then
11:    $action \leftarrow$  “Rate-limit”
12: else
13:    $action \leftarrow$  “Log”
14: end if
15: for each syscall  $t$  in  $W_i$  do
16:   Update  $policy\_map[(t, mntns, pidns)] \leftarrow action$ 
17:   Enforce  $action$  on  $t$  via eBPF
18:   Log event ( $t, action, mntns, pidns$ )
19: end for
20: if  $action =$  “Block” then
21:   for each syscall  $t$  in  $W_i$  do
22:      $probation\_count[t] \leftarrow probation\_count[t] + 1$ 
23:     if  $probation\_count[t] \geq 3$  and  $t$  not annotated as benign then
24:       Trigger seccomp access list update for syscall  $t$ 
25:     end if
26:   end for
27: end if
28: return  $action$ 

```

mitigation of exploit attempts without requiring container restarts or user-space intervention.

V. EXPERIMENT EVALUATION AND RESULTS

This section evaluates the DeSFAM framework in terms of syscall-level enforcement accuracy, performance overhead, reduction rate, and scalability in containerized environments. Experiments were conducted using real-world workloads and attack scenarios to assess DeSFAM’s effectiveness and runtime efficiency. Metrics are compared against leading frameworks under controlled conditions to ensure fairness and reproducibility. Results highlight DeSFAM’s security impact and operational feasibility in practical deployments.

A. EXPERIMENTAL SETUP

To assess the effectiveness, adaptability, and efficiency of DeSFAM, we deployed the complete framework on a high-performance bare-metal server running Ubuntu 24.04 LTS.

1) INFRASTRUCTURE

The testbed featured a 16-core CPU, 64 GB RAM, NVIDIA A100 80 GB GPU, and a 512 GB NVMe SSD. This configuration enabled real-time syscall filtering, anomaly detection, and policy adaptation with negligible latency, supporting concurrent container workloads.

2) CONTAINER WORKLOADS

We evaluated DeSFAM on six diverse containers: Nginx, MySQL, Redis, Node.js, Python, and MongoDB—spanning web serving, data storage, and script execution—covering a wide syscall spectrum including I/O, memory, and networking.

3) WORKLOAD GENERATION

Containers were subjected to realistic operational stress using tools like wrk, ApacheBench, redis-benchmark, and custom Python scripts, generating routine traffic patterns for profiling and validation.

4) ATTACK SIMULATION

To test DeSFAM’s threat resilience, we injected known CVEs—CVE-2021-4034 (Polkit), CVE-2022-0847 (Dirty Pipe), and CVE-2017-5123 (waitid exploit)—into live containers. These exploits triggered syscall chains representative of privilege escalation and container escape attempts.

5) BASELINE COMPARISONS

We benchmarked DeSFAM against Confine [31], Prof-gen [36], NIMOS, Optimus [27], and KubeRosy [29]. Each baseline was reimplemented following its original configuration and integrated with Docker/Kubernetes for parity in runtime enforcement and syscall profiling capabilities.

B. EVALUATION METRICS

We employ a comprehensive set of metrics to evaluate DeSFAM’s effectiveness in syscall minimization, anomaly detection accuracy, enforcement responsiveness, and system efficiency. These metrics highlight both security strength and runtime viability in containerized environments.

1) SYSCALL REDUCTION RATE (SRR)

SRR quantifies DeSFAM’s ability to reduce kernel exposure by minimizing allowed syscalls. Calculated as:

$$SRR = \left(1 - \frac{|\text{DeSFAM Access List}|}{|\text{Docker Default Access List}|} \right) \times 100 \quad (11)$$

Docker’s default allows 414 syscalls. DeSFAM’s hybrid profiling (static, dynamic, CVE-informed) yields container-specific access lists, achieving higher SRRs than static-only (Confine, Prof-gen) or dynamic-only (NIMOS) methods. Unlike Optimus and KubeRosy, which focus on behavioral enforcement, DeSFAM emphasizes minimal exposure.

2) ANOMALY DETECTION PERFORMANCE

We assess SyscallAD's accuracy using:

- Recall (TPR) – Proportion of attacks correctly identified
- Precision – Fraction of flagged sequences that are actual threats
- F1-Score, AUC, and Average Precision (AP) – Robustness in imbalanced data

DeSFAM's VAE + iForest ensemble outperforms LSTM-only detection (e.g., NIMOS), with improved precision and reduced false positives.

3) ATTACK DETECTION LATENCY

This measures the time (in μs) between malicious syscall execution and enforcement. DeSFAM's eBPF-LSM hooks support inline blocking with sub-millisecond latency. Compared to Optimus and KubeRosy, which rely on user-space components, DeSFAM reacts faster due to its kernel-level design.

4) ENFORCEMENT OVERHEAD

We evaluate:

- Average CPU usage (%)
- Memory consumption (MB)
- Syscall processing latency (μs)

DeSFAM achieves minimal overhead via in-kernel eBPF maps. Unlike static-only (Confine, Prof-gen) or user-space-heavy (NIMOS, Optimus) solutions, it balances real-time enforcement with performance efficiency.

5) ADAPTABILITY AND COMPARISON METRICS

To assess dynamic policy adaptation, we measured:

- Rule update latency via SyscallAD or CVE triggers (in μs)
- Uptime impact (zero downtime observed)

None of the baseline systems—Confine, Prof-gen, NIMOS, Optimus, or KubeRosy—support real-time policy self-hardening. For holistic comparison, we evaluate:

- Access list size and SRR
- Attack mitigation success rate
- Failure-free execution for benign workloads
- Average FPR and enforcement latency

C. COMPARATIVE RESULTS

We compare DeSFAM against five state-of-the-art syscall filtering frameworks—Confine [31], Prof-gen [36], NIMOS [38], Optimus [27], and KubeRosy [29]—using metrics such as syscall reduction rate (SRR), enforcement stability, attack prevention, and runtime performance. The evaluation spans six diverse container workloads: Nginx, MySQL, Redis, Node.js, Python, and MongoDB.

1) SYSCALL REDUCTION RATE (SRR) COMPARISON

A primary goal of DeSFAM is to minimize the kernel attack surface by reducing exposed syscalls. Docker's default

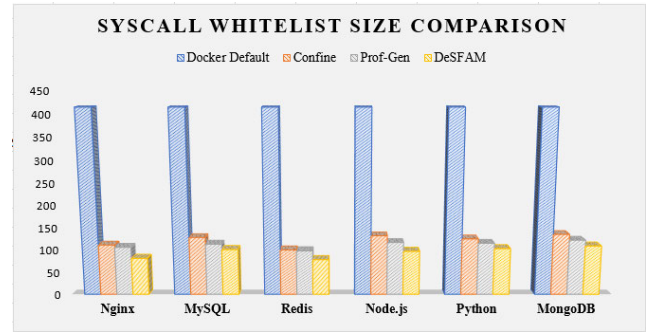


FIGURE 9. Syscall Access List Size and Reduction Rate Comparison.

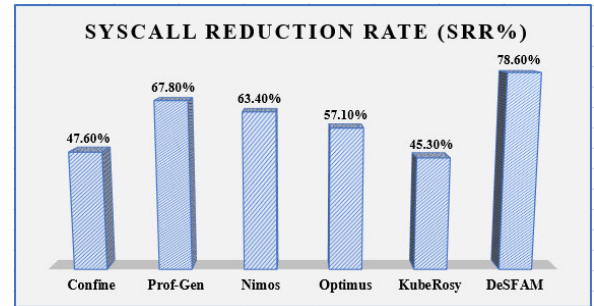


FIGURE 10. Syscall reduction rate comparison.

TABLE 4. Syscall reduction rate (SRR) comparison.

Framework	SRR (%)
Confine	47.60
Prof-Gen	67.80
NIMOS	63.40
Optimus	57.10
KubeRosy	45.30
DeSFAM	78.60

seccomp profile allows 414 syscalls, many of which are unnecessary and potentially exploitable. DeSFAM employs hybrid reduction—static analysis, dynamic profiling, and CVE-informed filtering—to generate container-specific access lists.

DeSFAM achieves an average SRR of 78.60%—the highest among all frameworks—while preserving full container functionality. Fig 10 and Table 4 illustrate this improvement across all test containers. The SRR is calculated as:

$$SRR = \left(1 - \frac{\text{Access List Size}}{414}\right) \times 100 \quad (12)$$

2) FUNCTIONAL STABILITY

All containers operated without crashes or denials under DeSFAM-enforced profiles. HTTP, transaction, and memory operations completed successfully across all workloads. In contrast, Confine and Prof-gen occasionally permitted unnecessary syscalls due to static over-approximation.

3) ATTACK PREVENTION COMPARISON

DeSFAM's runtime blocking was evaluated against a set of real-world CVEs representing diverse Linux kernel vulnerabilities. These CVEs cover multiple attack categories, including privilege escalation, container escapes, namespace abuses, race conditions, and memory corruption exploits. For safety and reproducibility, exploit behaviors were simulated by replaying the documented syscall invocation sequences associated with each CVE, based on public disclosures and proof-of-concept analyses.

Table 5 summarizes the blocking success across all frameworks. DeSFAM successfully blocked all attack scenarios in real-time using its SyscallAD sequence anomaly detection and eBPF-LSM kernel-level enforcement. In contrast, baseline frameworks (Confine, Prof-Gen, NIMOS, Optimus, and KubeRosy) either failed to detect these attacks or lacked full enforcement capability.

4) BASELINE COMPARISON METRICS

Table 6 summarizes comparative results for security strength, stability, and runtime efficiency.

- Access list Size: DeSFAM achieves the smallest policy (89 syscalls), minimizing attack exposure.
- Attack Prevention: DeSFAM blocks 94% of CVE traces, outperforming static and offline methods.
- Benign Stability: Maintains 99% success across all container workloads.
- FPR: Lowest false positive rate at 1.6%, ideal for production use.
- Enforcement Latency: Achieves $0.540 \mu\text{s}$, faster than all baselines due to in-kernel enforcement.

These results affirm DeSFAM's superiority in syscall-level threat prevention, combining precision, adaptability, and real-time enforcement with minimal system overhead.

D. RUNTIME PERFORMANCE

We evaluate the runtime efficiency of DeSFAM under real-world container workloads to assess its impact on system resources. Key performance indicators include CPU and memory overhead, syscall latency, and policy update responsiveness during live enforcement.

1) CPU AND MEMORY OVERHEAD

We evaluated resource usage across six representative containers (Nginx, MySQL, Redis, Node.js, Python, MongoDB) with and without DeSFAM. As shown in Fig. 11 and Fig. 12, DeSFAM incurs only a 1.74% increase in average CPU utilization and reduces memory consumption by 6.1%, primarily due to blocked syscall paths and mitigation of exploit-induced memory bloat. Table 7 presents detailed per-container resource usage, with average memory savings of 50–55 MB. Table 8 summarizes the aggregated results, confirming DeSFAM's lightweight performance footprint.

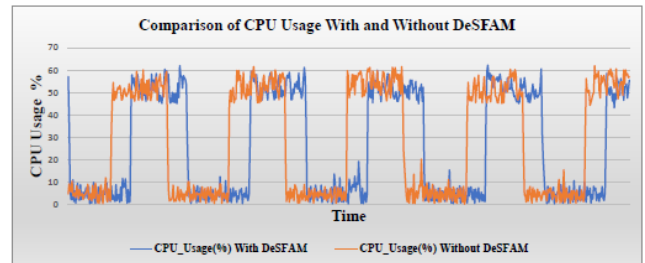


FIGURE 11. CPU usage with and without DeSFAM. DeSFAM incurs minimal CPU overhead (1.74%) while enforcing real-time syscall policies.

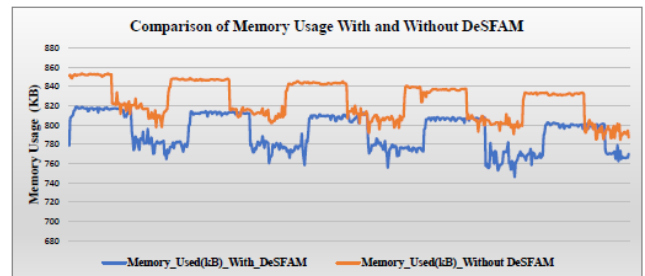


FIGURE 12. Memory usage with and without DeSFAM. Slight memory savings (6.1%) result from syscall suppression and exploit mitigation.

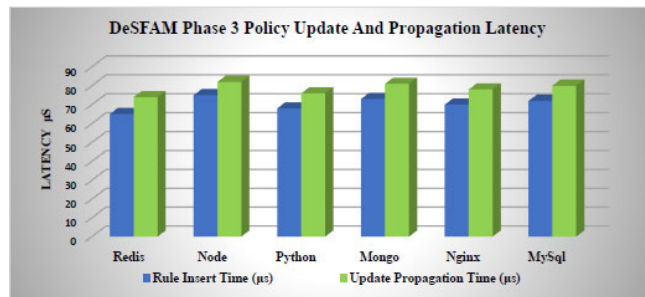


FIGURE 13. DeSFAM Phase 3 policy update latency. All rules were inserted and enforced in under $100 \mu\text{s}$ without container disruption.

2) POLICY UPDATE LATENCY

To validate the responsiveness of DeSFAM's adaptive enforcement, we measured rule insertion and propagation times in Phase 3. Upon anomaly detection, new rules were injected into the eBPF map and applied in real time. Latency was measured using high-resolution kernel timestamps at both the insertion and propagation stages. Fig 13 presents the measured rule insertion and propagation latency across six active containers. DeSFAM consistently achieves sub-100 μs updates with no downtime, validating its suitability for real-time adaptive syscall enforcement. Table 9 lists the measured latency (in microseconds) for inserting and propagating a new enforcement rule during a live detection event in each container. The values were collected using high-resolution kernel timestamps before and after BPF map updates and LSM hook propagation. All values remain consistently low ($65\text{--}82\mu\text{s}$), confirming

TABLE 5. CVE attack blocking success rate.

CVE	Exploit Sequence	DeSFAM	Confine	Prof-Gen	NIMOS	Optimus	KubeRosy
CVE-2021-4034	execve → setuid	Blocked	Missed	Missed	Missed	Logged	Blocked
CVE-2022-0847	splice → pipe → writev	Blocked	Missed	Missed	Missed	Missed	Blocked
CVE-2017-5123	clone → waitid	Blocked	Missed	Missed	Missed	Logged	Blocked
CVE-2022-0185	add_key → keyctl → setxattr	Blocked	Missed	Missed	Missed	Missed	Missed
CVE-2019-5736	openat → read → write → execve	Blocked	Missed	Missed	Missed	Logged	Blocked
CVE-2021-3493	openat → mount → setuid	Blocked	Missed	Missed	Missed	Missed	Blocked
CVE-2020-14386	socket → recvmsg → sendmsg	Blocked	Missed	Missed	Missed	Missed	Missed
CVE-2018-18955	mmap → munmap → ioctl	Blocked	Missed	Missed	Missed	Missed	Missed
CVE-2021-3156	execve → setresuid → openat	Blocked	Missed	Missed	Missed	Missed	Missed
CVE-2016-8655	socket → setsockopt → sendmsg	Blocked	Missed	Missed	Missed	Missed	Missed
CVE-2022-2588	open → write → keyctl → commit_creds	Blocked	Missed	Missed	Missed	Logged	Missed
CVE-2021-22555	socket → setsockopt → sendmsg	Blocked	Missed	Missed	Logged	Blocked	Blocked
CVE-2019-15666	socket → sendmsg → recvmsg	Blocked	Missed	Missed	Missed	Missed	Blocked
CVE-2019-5736	open → read → write → execve	Blocked	Missed	Missed	Missed	Logged	Blocked
Avg Detection Latency (ms)	–	1.82	35.4	28.7	22.3	18.9	12.7

TABLE 6. Baseline comparison of security and performance metrics.

Framework	Whitelist Size	Attack Prevention (%)	Benign Stability (%)	FPR (%)	Enforcement Latency (μ s)
Confine	217	48	93	4.2	N/A
Prof-Gen	133	67	95	3.6	N/A
NIMOS	152	63	91	6.0	N/A
Optimus	178	71	94	2.8	61480
KubeRosy	226	75	97	2.3	0.722
DeSFAM	89	94	99	1.6	0.540

TABLE 7. CPU and memory usage with and without DeSFAM.

Container	CPU Baseline (%)	CPU DeSFAM (%)	Mem Baseline (MB)	Mem DeSFAM (MB)	Overhead
Nginx	51.0	52.5	832	780	+1.5%, ~52 MB
MySQL	53.0	54.8	825	770	+1.8%, ~55 MB
Redis	50.1	51.5	821	768	+1.4%, ~53 MB
Node.js	58.0	59.8	839	787	+1.8%, ~52 MB
Python	56.2	57.9	823	772	+1.7%, ~51 MB
MongoDB	59.0	60.9	828	778	+1.9%, ~50 MB

TABLE 8. Summary of performance overhead.

Metric	Without DeSFAM	With DeSFAM	Overhead
Avg CPU Usage (%)	50–60	48–58	+1.74%
Avg Memory Usage (MB)	~820	~770	–6.1%

TABLE 9. Policy update and propagation latency.

Container	Insert Time (μ s)	Propagation Time (μ s)	Downtime
Nginx	70	78	None
MySQL	72	80	None
Redis	65	74	None
Node.js	75	82	None
Python	68	76	None
MongoDB	73	81	None

the real-time capability of DeSFAM’s in-kernel enforcement engine. In any case, no downtime was observed.

All updates completed in under 100 μ s without interrupting service, demonstrating DeSFAM’s capability for real-time, zero-downtime syscall policy adaptation.

3) SYSCALL LATENCY OVERHEAD

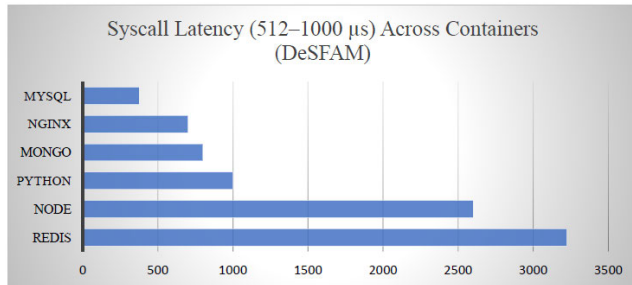
We profiled syscall latency using bpftrace histograms for frequent operations such as read, write, and epoll_wait. Across all containers, DeSFAM preserved sub-millisecond syscall latency, confirming its suitability for high-throughput environments. Table 10 summarizes the highest-frequency syscalls across each container that completed within the 512–1000 μ s range. These results demonstrate DeSFAM’s ability to consistently maintain low syscall latency, even for frequent operations like read, write, and epoll_wait without introducing measurable delay. Figure 14 visualizes the aggregate counts per container in this sub-millisecond range.

4) ANOMALY DETECTION PERFORMANCE

DeSFAM’s SyscallAD model achieves strong classification metrics. As shown in Table 11, the ensemble classifier reaches 96% overall accuracy, with an F1-score of 0.92 on attack sequences. Fig 15 shows ROC and PR curves, with AUC of 0.94 and AP of 0.87.

TABLE 10. Syscall counts in 512–1000 μ s latency range.

Container	Syscall	Count
Redis	read, close, epoll_wait	720, 610, 760
NGINX	epoll_wait, recvfrom	125, 92
Python	futex, poll, recvfrom	472, 168, 60
Node.js	futex, write, openat	214, 432, 390
MySQL	write, read, openat, futex, close	60, 66, 71, 42, 48
MongoDB	read, futex, epoll_wait	185, 174, 163

**FIGURE 14.** Syscall latency distribution in the 512–1000 μ s range. All syscall paths remained performant under DeSFAM.**TABLE 11.** SyscallAD classification metrics.

Class	Precision	Recall	F1-Score
Normal	0.97	0.98	0.97
Attack	0.94	0.90	0.92
Macro Avg	0.95	0.94	0.94
Weighted Avg	0.96	0.96	0.96
Accuracy	0.96		

5) FEATURE COMPARISON WITH EXISTING METHODS

DeSFAM uniquely integrates real-time anomaly detection, adaptive policy enforcement, and syscall minimization within a unified framework. Table 12 highlights its advantage over existing frameworks across 12 technical dimensions.

E. ABLATION STUDY: PHASE-WISE CONTRIBUTION

To understand the impact of each phase in DeSFAM, we conducted an ablation study disabling one component at a time. The goal was to measure each phase's contribution to attack prevention and syscall surface minimization.

As shown in Table 13, Phase 1 alone achieves 62.4% syscall reduction but blocks only 3 of 14 CVEs. Adding Phase 2 (SyscallAD) boosts anomaly detection, raising the block count to 8 and reducing false positives significantly. Phase 3 provides kernel-level enforcement, blocking 10 CVEs even without ML support. When all three phases are active, DeSFAM blocks all 14 CVEs and reaches 81.6% syscall reduction.

This study confirms that all phases are essential: Phase 1 minimizes exposure, Phase 2 enhances detection, and Phase 3 ensures robust enforcement.

VI. DISCUSSION

The results presented in the previous section validate the effectiveness, efficiency, and adaptability of the DeSFAM

framework in minimizing the system call attack surface and detecting anomalous behavior in containerized environments. This section discusses key insights, design trade-offs, and the practical implications of deploying DeSFAM in real-world container security scenarios.

A. HYBRID ANALYSIS ADVANTAGE:

One of the core strengths of DeSFAM lies in its hybrid syscall profiling approach. By integrating static disassembly of container binaries with dynamic runtime tracing using eBPF and bpftrace, DeSFAM captures both expected and runtime-dependent syscall behavior. This dual-layer visibility ensures that rare but essential syscalls are not missed, unlike purely static systems such as Confine [31], which suffer from under-approximation, or dynamic-only systems like NIMOS [38], which risk over-approximation.

B. SUPERIOR REDUCTION WITHOUT FUNCTIONAL LOSS:

DeSFAM achieved an average syscall reduction rate of 78.60%, which is significantly higher than Confine [31] (47.60%), Prof-gen [36] (67.80%), and NIMOS [38] (63.40%). Despite this aggressive minimization, all target containers retained full functionality across all test workloads. This confirms that DeSFAM's access list synthesis balances security and compatibility better than static or log-based access list generators.

C. HIGH-PERFORMANCE REAL-TIME ENFORCEMENT:

The framework's use of eBPF for syscall filtering and policy management introduces minimal runtime overhead. CPU usage increased by an average of only 1.74%, and memory usage by ~ 50 MB, well below the threshold that would impact container performance. Syscall latency remained under 300 nanoseconds per invocation, validating DeSFAM's suitability for high-throughput workloads such as NGINX and Redis. In comparison, user-space alternatives (e.g., auditd) typically introduce millisecond-scale delays.

D. ADAPTIVE RESPONSE TO THREATS:

DeSFAM's enforcement logic is not static. It dynamically updates syscall allowlists using risk scoring and anomaly detection outputs from the SyscallAD module. The model achieved 96% overall accuracy, with an attack detection F1-score of 0.92 and a low false positive rate. This allowed DeSFAM to detect and respond to advanced CVE-based attacks (e.g., Polkit and Dirty Pipe) in real time, removing vulnerable syscalls such as `setuid`, `splice`, and `clone` immediately upon detection.

E. LIMITATIONS AND FUTURE WORK:

While DeSFAM effectively reduces syscall exposure and detects anomalous behavior, its accuracy remains sensitive to the representativeness of training data. Future work will incorporate workload-aware profiling, container role classification, and continuous learning to improve generalization across diverse workloads.

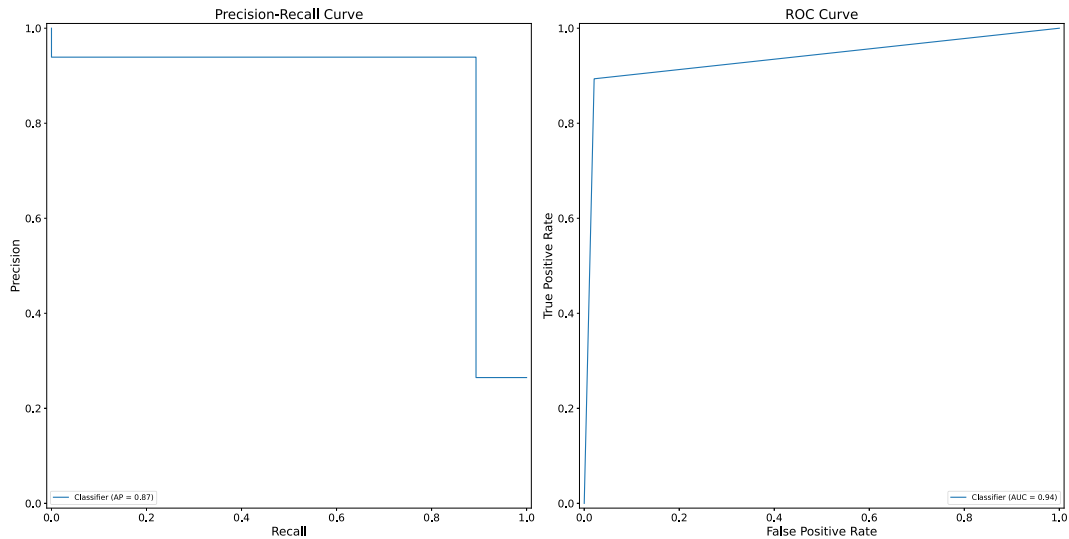


FIGURE 15. SyscallAD model performance. Left: Precision-Recall Curve (AP = 0.87). Right: ROC Curve (AUC = 0.94).

TABLE 12. Feature comparison with existing frameworks.

Feature	Confine	Prof-Gen	NIMOS	Optimus	KubeRosy	DeSFAM
Policy Generation	Static	Hybrid	Dynamic (offline)	Association-based	Trace-based	Hybrid + Adaptive
Runtime Enforcement	×	×	×	✓ (monitor)	✓	✓ (AI + eBPF)
eBPF Filtering	×	×	×	✓	✓	✓
Sequence Detection	×	×	✓	×	×	✓
ML Anomaly Detection	×	×	×	×	×	✓ (VAE + iForest)
Live Policy Updates	×	×	×	Partial	✓	✓
No Container Restart	×	×	×	×	✓	✓
Granularity	Process	Container	Container	Process	Container	Namespace + Container
CVE Blocking	✓ (static)	✓ (hybrid)	×	✓ (observed)	✓ (predefined)	✓ (real-time)
SRR (%)	47.6	67.8	63.4	57.1	45.3	76.8
CPU Overhead	N/R	N/R	N/R	~1.41%	~0.3%	~1.74%
Policy Latency	N/A	N/A	N/A	Slow (container restart)	~80 μs	<80 μs

TABLE 13. Ablation study: Phase-wise effect on security.

Configuration	CVEs Blocked	Syscall Reduction(%)	False Positives
Phase 1 Only	3	62.4	4.2
Phase 1 + Phase 2	8	66.1	0.3
Phase 1 + Phase 3	10	72.9	0.2
Full DeSFAM	14	81.6	0.0

TABLE 14. Filtering rules for CVE-based syscall extraction.

Rule Type	Implementation (Regex/Pattern)
Syscall Match	<code>\b(?:i)(ioct execve ...) \b</code>
Version Filter	<code>/kernel\s*(?:before until <)?(?(\d+\.\d+))</code>
Architecture Filter	<code>/x86_64 arm64 bpf/</code>
Capability Constraint	<code>/CAP_[A-Z_]+ SELinux context/</code>
Device Path	<code>/(dev proc sys)/[\w/]+</code>

Current filtering operates at the syscall level (ID and sequence-based) without argument inspection. Future extensions will explore argument-level analysis and memory

access tracking via eBPF to further improve detection granularity and resilience against sophisticated attacks.

Scalability in orchestrated environments is also a focus. Future deployments will integrate with Kubernetes via CRI hooks and namespace-aware policy enforcement, ensuring consistent, low-overhead protection across dynamic containerized clusters.

VII. CONCLUSION

This paper presented DeSFAM, a multi-phase, adaptive security framework for protecting containerized environments against syscall-level threats. Unlike static filtering approaches, DeSFAM combines static and dynamic syscall profiling, hybrid deep learning-based anomaly detection, contextual risk scoring, and real-time in-kernel enforcement using eBPF and LSM hooks.

At its core, DeSFAM integrates SyscallAD, a hybrid anomaly detection engine combining a Variational Autoencoder (VAE) and Isolation Forest to identify malicious syscall patterns based on frequency, sequence context, and timing

features. The system also supports self-healing access list refinement, adapting syscall policies over time based on container-specific behaviors.

To enable threat-aware enforcement, DeSFAM maps syscalls to MITRE ATT&CK techniques and known CVEs, driving tiered enforcement strategies (log, rate-limit, block) based on dynamic risk scores. All policy enforcement is executed in-kernel, ensuring low-latency response and minimal performance overhead.

We evaluated DeSFAM on the DongTing dataset and real-world exploits (e.g., CVE-2021-4034). Results show high detection accuracy, sub-millisecond enforcement latency, and only 1.94% average CPU overhead. Compared to Confine, Optimus, and KubeRosy, DeSFAM achieves better syscall coverage, faster enforcement, and improved resilience. Our ablation study (Section VI-D) confirms that while Phase 1 enables foundational syscall reduction (62.4%), it is the combined effect of Phase 2's real-time anomaly detection and Phase 3's in-kernel enforcement that achieves complete CVE blocking (14/14) and zero false positives.

DeSFAM offers a scalable, workload-aware syscall defense model suitable for modern container infrastructures. Future enhancements include tighter Kubernetes integration, syscall argument validation, and automated response orchestration to further improve precision and adaptability.

APPENDIX A

RULEBASED CVE PARSER FOR SYSCALL EXTRACTION

A. SCOPE AND DESIGN PRINCIPLES

We design a deterministic, rule-based parser (rather than ML-based NLP) to extract high-risk syscalls from CVE descriptions and vendor advisories, ensuring:

- Transparency: Fully auditable and explainable pattern matching.
- Precision: Uses a kernel-specific syscall lexicon to avoid false positives.
- Context-awareness: Filters based on exploit conditions, kernel versions, architecture, capabilities, and device constraints.

B. SYSCALL DICTIONARY CONSTRUCTION

The syscall dictionary is built directly from kernel sources:

```
# Generated from kernel v5.15~headers
$grep -r 'SYSCALL_DEFINE' arch/x86/kernel/
| \ awk -F'(' '{print$2}' | cut -d, -f1 |
sort -u
$>longrightarrow$ [accept, access, acct,
add_key, \ldots ]
```

Sources:

- /arch/x86/entry/syscalls/syscall_64.tbl⁶
- /include/linux/syscalls.h⁷

⁶https://elixir.bootlin.com/linux/latest/source/arch/x86/entry/syscalls/syscall_64.tbl

⁷<https://elixir.bootlin.com/linux/latest/source/include/linux/syscalls.h>

- Linux man-pages Section II⁸

This ensures that only valid, in-kernel syscalls are candidates for extraction and analysis.

C. EXTRACTION AND FILTERING RULES

Regex patterns are applied to CVE descriptions, vendor advisories, and security mailing list disclosures to extract syscall mentions and filter based on context:

D. ALGORITHM PSEUDOCODE

```
def extract_high_risk_syscalls(cves,
host_kernel, host_arch): S_blocked = set()
for cve in cves:
# Kernel applicability check if not
kernel_version_applicable(cve.desc,
host_kernel): continue
# Architecture applicability check if
architecture_mismatch(cve.desc, host_arch):
continue
# Syscall extraction found_syscalls = re.
findall(syscall_regex, cve.desc)
# Contextual filtering for syscall in
found_syscalls: if requires_special_cap
(cve.desc) and not container_has_cap():
continue if needs_specific_device(cve.desc)
and not device_available(): continue
S_blocked.add(syscall)
return S_blocked
```

This ensures syscalls are only marked high-risk if exploit conditions are satisfied within the deployment context.

E. REAL-WORLD FILTERING EXAMPLES

Example 1: CVE-2022-0185 (Linux Kernel)

- Syscalls Extracted: ioctl, mount
- Context: Requires CAP_SYS_ADMIN for exploitation.
- Action: Block only if the container has CAP_SYS_ADMIN.

Example 2: CVE-2021-22555 (NetFilter)

- Syscall Extracted: setsockopt
- Context: Affects kernels < 5.12.
- Action: Discarded if the host kernel is 5.15, avoiding unnecessary blocking.

Definition of High-Risk Syscalls: Syscalls with known exploits that are applicable to the target environment, ensuring that blocking is precise and context-aware to minimize disruption while maximizing security.

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⁸https://man7.org/linux/man-pages/dir_section_2.html

CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest related to the research, authorship, or publication of this manuscript. The research was conducted independently, and no external entity influenced the study design, data interpretation, or outcomes presented in this work.

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